

Name: Abhay Gami

Sem: 7

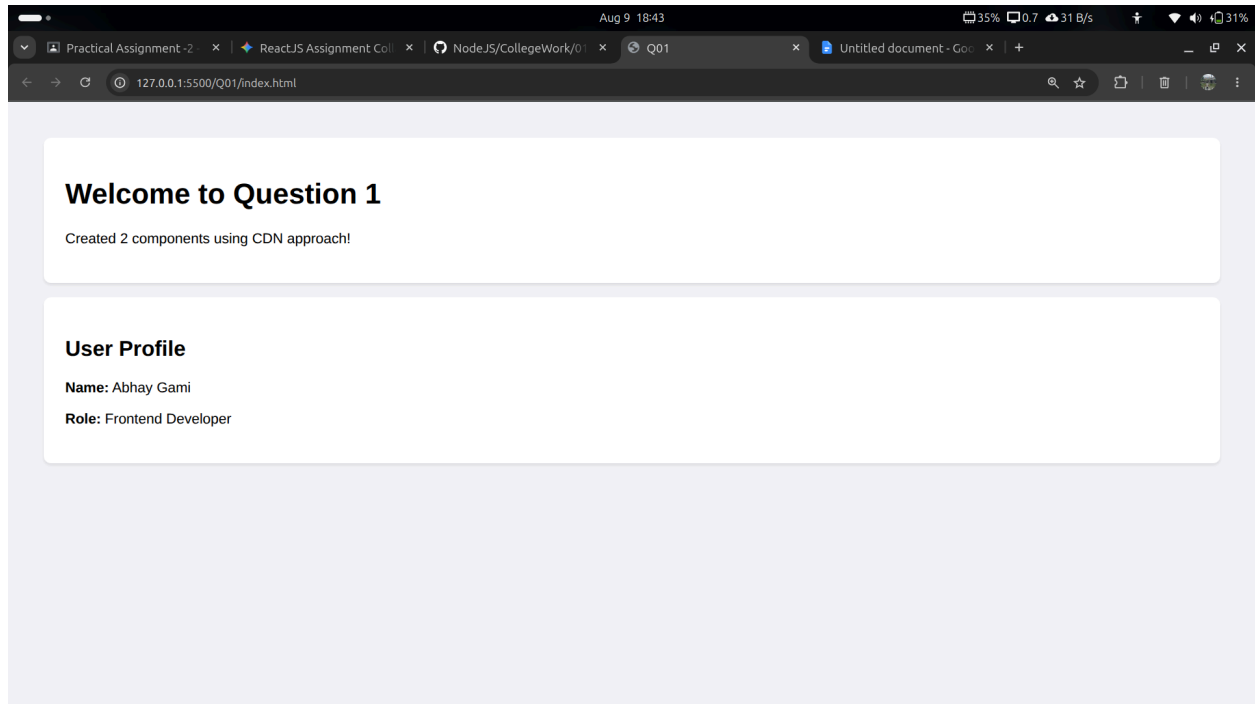
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Div: A

**Sub: Web Development using fullstack
(701)**

Practical Assignment 2

Question 1: CDN-Based React Application

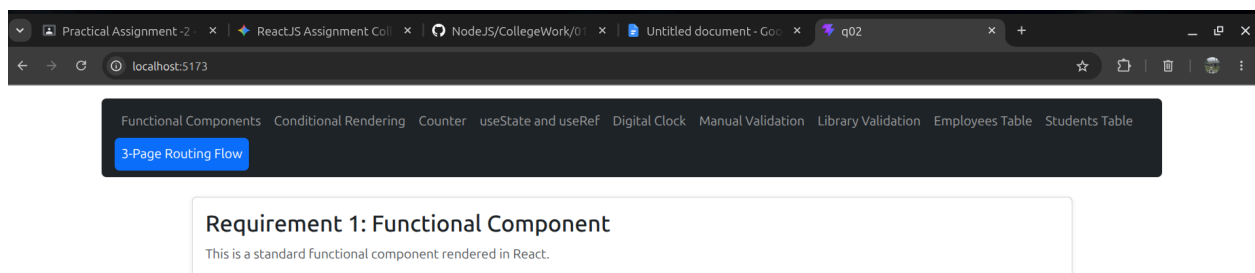


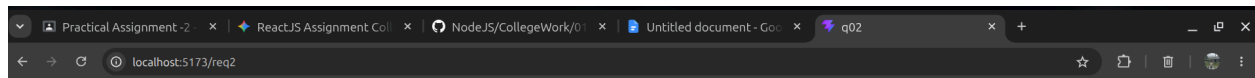
Question 2

Setup a React application using **Vite**, integrate **Bootstrap** for styling, and set up **React Router** to navigate across components satisfying the following requirements:

Module Requirements:

1. **Functional Component:** Create and display a basic functional component.
2. **Component Features & Architecture:** Implement a scenario demonstrating:
 - Conditional Rendering
 - List Rendering (using `.map()`)
 - Nested Components
 - Component Containment (using `props.children`)
3. **State Management (Counter):** Develop a counter component with **Increment**, **Decrement**, and **Reset** functionality.
4. **Hooks (`useState` & `useRef`):** Create a component using `useState` for state tracking and `useRef` for directly manipulating or focusing DOM elements.
5. **Lifecycle Management (Digital Clock):** Build a live digital clock utilizing `useState` to store the current time and `useEffect` with clean-up intervals.
6. **Manual Form Validation:** Implement a form with real-time (live) manual validation for email address and password criteria.
7. **Library Form Validation:** Implement live form validation using a third-party validation library (e.g., `react-hook-form`).
8. **JSON Data Fetching (Employees):** Import/fetch data from a local `employees.json` file and render it inside a styled Bootstrap table.
9. **Data Searching & Filtering (Students):** Import/fetch data from `students.json` and display it in a table. Include:
 - **Search bar:** Filter dynamically based on the student's **first name**.
 - **Filter dropdowns:** Filter by **Semester**, **Division**, and **Gender**.





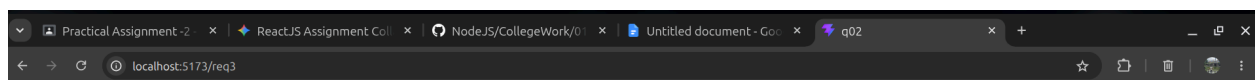
Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 2: Conditional, List, Nested & Children Demo

User Activity List

Alice	Online
Bob	Offline
Charlie	Online



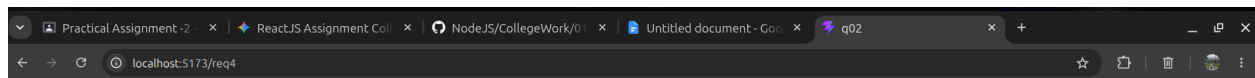
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3-Page Routing Flow

Requirement 3: Counter Component

Count: 3

Increment Decrement Reset



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3-Page Routing Flow

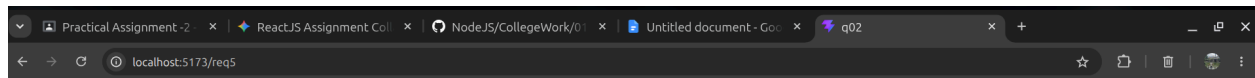
Requirement 4: useState & useRef

Type something (useState):

aaa

State output: aaa

Focus Input via useRef

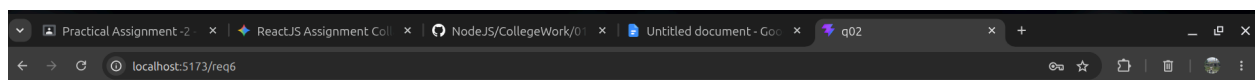


Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 5: Digital Clock

6:46:17 PM



Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 6: Live Manual Validation

Email Address

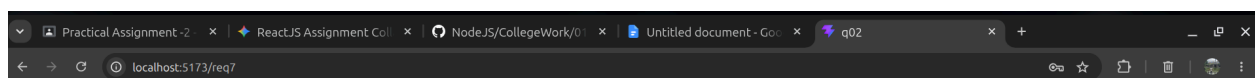
abha

Enter a valid email address (e.g. name@domain.com)

Password

..

Password must have min 6 chars, 1 uppercase, 1 lowercase, 1 digit & 1 special char (@\$!%*?&)



Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 7: Live Validation

Email Address

info@abhay.io

Password

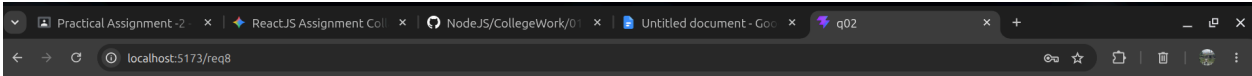
Submit

localhost:5173 says

Form Submitted Successfully!

```
{
  "email": "info@abhay.io",
  "password": "Abhay@098"
}
```

OK

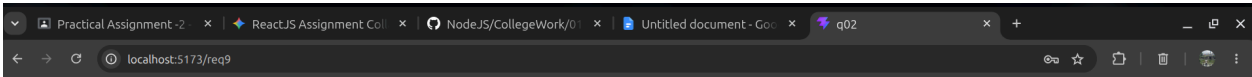


Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 8: Fetch Employees JSON

ID	Name	Department	Salary (₹)
101	Rahul Sharma	Engineering	75000
102	Priya Patel	Design	68000
103	Amit Verma	Marketing	62000
104	Neha Gupta	HR	58000



Functional Components Conditional Rendering Counter useState and useRef Digital Clock Manual Validation Library Validation Employees Table Students Table

3-Page Routing Flow

Requirement 9: Fetch Students JSON with Search & Filters

All Semesters

All Divisions

All Genders

ID	First Name	Last Name	Semester	Div	Gender
1	Aarav	Shah	Sem 4	A	Male
2	Ananya	Mehta	Sem 6	B	Female
3	Rohan	Joshi	Sem 4	B	Male
4	Diya	Trivedi	Sem 6	A	Female

Question 3: Dynamic "To-Do" Application

